MIL-A-25175C(USAF)
6 February 1976
Superseding
MIL-A-24175B(USAF)
4 February 1971

MILITARY SPECIFICATION

AIR TRANSPORT, NONTACTICAL, PACKING FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers packing procedures for preparation of material, equipment, and supplies for nontactical air transport by military and commercial aircraft.

1.2 The importance in selecting containers for air cargo is to attain the minimum tare and cube consistent with the protection required to withstand the hazards encountered in air transport.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

FSC PACK

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Joint Military

Preparation of Hazardous Materials for Military Air Shipments (AFR 71-4/ DSAM 4145.3/TM 38-250/NAVSUP PUB 505/MCO P4030.19).

Department of Transportation

Title 14 Code of Federal Regulations Part 103

2.3 Source of Documents

a. Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

b. Applications for copies of DOT regulations should be addressed to the Superintendent of Documents, Washington DC 20402.

3. REQUIREMENTS

3.1 Materials. Materials shall conform to applicable specifications as specified herein. Authorization to use materials that are not covered by specifications or specifically described herein shall be obtained from the procuring activity.

3.2 Weight and Size Limitations. All containers shall be of minimum weight and cube consistent with the weight and cube limitations of the applicable container specification without compromising the protection required. Shipping containers of identical items and quantities shall be of uniform size and shape. Shipping containers with a gross weight of 200 pounds or over will be provided with longitudinal skids of sufficient size and attached in a manner to permit the use of lifting devices. Containers, 60 inches or more in length or width, will be provided with four-way entry whenever practical. Load-bearing members will be provided in areas that are in contact with lifting devices.

3.3 Disassembly of Items. All items that will reduce the cubic displacement by disassembly, shall be disassembled, if practicable. Whenever disassembly of items alter the calibrations or settings that will effect the operation of the item or its specific functions, or if special tools are required to re-assemble the item, disassembly is not permissible unless specified or directed by the procuring activity. For hazardous materials see 3.6.

3.4 Use of Exterior Containers. Exterior containers or pallets shall be used to assemble multiple units of items or intermediate packages into a unit load or a protective enclosure. Irregular shaped items shall be fastened, blocked and braced to prevent free movement within the container

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or on a pallet, except where movement is controlled by adjacent cushioning media. Type 3 loads shall be cushioned, blocked, braced, fastened or otherwise secured in accordance with MIL-STD-1186. Blocking and bracing must be sufficient to transfer the tie down forces through the container to the item. For hazardous materials see 3.6.

3.5 Selection of Containers and Pallets. Selection of containers or pallets shall be based on the weight and cube limitations set forth in the container specification or standard as applicable. When the weight and cube of an item or items permit the use of more than one container, it shall be the option of the contractor to select any container in MIL-STD-794 that will provide adequate protection for air shipment. Palletized loads shall not exceed the weight and size limitations of MIL-STD-147.

3.6 Hazardous Materials Shipped by Military Air. Hazardous materials shipped by military air or delivered to an aerial port of embarkation for military air shipment will be prepared for shipment in accordance with AFR 71-4/DSAM 4145.3/TM 38-250/NAVSUP PUB 505/MCO P4030.19, Preparation of Hazardous Materials for Military Air Shipments.

3.6.1 Hazardous Materials Shipped by Other Than Military Air. Hazardous materials shipped by air, other than military, will be prepared for shipment in accordance with the current requirements of the Civil Air Regulations (Air Transport Restricted Articles Tariff 6) as promulgated by the Code of Federal Regulations Title 14, Part 103.

3.6.2 Restrictions on Hazardous Materials Shipped by Air. Hazardous materials preserved and packaged in accordance with the requirements set forth in paragraphs 3.6 or 3.6.1 shall not be stripped or consolidated with other materials.

3.7 Closure and Strapping. Containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Strapping shall conform to QQ-S-781, PPP-S-760 or PPP-T-97.

3.8 Palletized Loads. When practicable, pallets shall be loaded and secured in accordance with MIL-STD-147. The maximum gross weight and size limitations of this standard shall not be exceeded.

3.9 Marking. All unit, intermediate, shipping containers and unboxed items shall be marked in accordance with MIL-STD-129. Pertinent precautionary markings necessary for full protection of the item shall be prominently located on the shipping container as specified in MIL-STD-129. After determination and identification, hazardous materials shall be marked and/or labeled in accordance with AFR 71-4. The location of these markings and/or labels shall be as specified in MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended Use. It is intended that this specification be used as a guide for the packer or repacker in selecting containers that will furnish adequate protection with the least possible tare weight. However, the selection of lightweight containers is not necessarily the solution to all air shipment problems. Metal containers may be used for air shipment of certain articles when necessary or when they are found to be economical for the purpose intended.

6.1.1 Packing methods and containers used for export shipments are designed to withstand superimposed loads, extreme handling conditions and long time exposure to all weather conditions. Frequently, out of necessity, these containers which will afford this sort of protection are extremely heavy. The weight of the containers and packaging materials are of great importance when transporting material by air.

6.1.2 Permissible Procedures for Air Shipments.

a. Strapping of Material. Ref. 30.1

b. Consolidation of Material. Ref 30.2

c. Use of a base, skids, pallet or cradle. Ref 30.3

d. Use of Lightweight Domestic Containers. Ref 30.4

e. Use of unit or intermediate pack. Ref 30.5

f. Selection of container for type of load. Ref 30.6

g. Removal of shipping container. Ref 30.7

h. Reinforcing of containers. Ref 30.8

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APPENDIX

10. SCOPE

10.1 This appendix provides general information pertaining to the preparation of material for air shipment.

20. APPLICABLE DOCUMENT

20.1 The following document, of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

Military

MIL-A-8421 Air Transportability Requirements, General Specification for

30. REFERENCES

30.1 Strapping of Material. Material prepared for air shipment may be shipped without an exterior container providing they fall into one of the following categories:

a. The material shall be of size, shape and weight that will permit manual or mechanical handling and tie down devices.

b. The material shall be rugged and shock resistant enough to withstand hazards of transportation, handling, and tie down equipment. This determination may be obtained from the engineering department.

c. When practicable, material may be stripped to the unit package for consolidation. Ref. 30.2.

d. Exterior container may be omitted on painted or preserved material that can withstand outdoor exposure for a limited time or material intended for outdoor use.

30.2 Consolidation of Material. It is permissible to consolidate material into a lightweight container providing packs can be stripped to the unit or intermediate package and weight and/or cube can be reduced without sacrificing the protection required.

30.3 Use of a base, skids, pallet or cradle. Material meeting the requirements of reference 30.1 but too heavy (over 200 pounds) or difficult to handle should be mounted on a base, skids, cradle, or pallet to provide for mechanical handling.

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30.4 Use of Lightweight Domestic Containers. Domestic containers will be used when substitution to preferred containers would not decrease the weight and cube. Material packed in metal containers shall not be repacked unless a substantial savings can be realized on weight and cubage and the protection remains adequate.

30.5 Use of the unit or intermediate pack. The unit or intermediate pack may be used as exterior containers providing the weight and cube of the container does not exceed the weight and cube limitations of the applicable container specification. Heavy exterior containers should be removed when unit or intermediate containers satisfy the requirement.

30.6 Selection of container for type of load. The type of load determines the container requirements. Types 1 or 2 loads require the inside dimensions of the container to be sufficiently exact so that the packed contents will fit snugly and give support to all faces of the box. Type 3 load does not fit the box snugly and may permit shifting. When permissible, Type 3 loads should be converted to Types 1 or 2 loads. Type 3 loads that can not be converted should be packed in accordance with paragraph 3.4.

30.7 Removal of shipping container. If the mode of transportation necessitates a heavier shipping container than necessary for air shipment, such containers should be so designed and marked to remove the heavy exterior container for transshipment by air. The container used for air shipment must be strong enough to withstand handling at the air terminal and the physical and environmental in-transit conditions aboard the aircraft. Guidance on the inflight physical and environmental conditions if needed can be obtained by referring to MIL-A-8421.

30.8 Reinforcing of containers. Containers may be reinforced in one or more of the following ways:

a. Steel strapping.

b. Filament tape.

c. Reinforced corners by nailed strapping.

d. Cleats.

e. Diagonals.

f. Fiberboard sleeves.

g. Fasteners.

h. Staples.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
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